

Dynamic Experts Synergy for Multi-Task Recommendation

Haotian Wu
Nanyang Technological University
Singapore
wu.haotian@ntu.edu.sg

Yingpeng Du*
Nanyang Technological University
Singapore
dyp1993@pku.edu.cn

Zhu Sun*
Singapore University of Technology
and Design
Singapore
sunzhuntu@gmail.com

Jie Zhang
Nanyang Technological University
Singapore
zhangj@ntu.edu.sg

Puay Siew Tan
Singapore Institute of Manufacturing
Technology, A*STAR
Singapore
pstan@simtech.a-star.edu.sg

Abstract

Expert-sharing patterns have emerged as promising paradigms for multi-task learning (MTL) in recommender systems, enabling efficient resource allocation and dynamic modeling of diverse tasks. In this paper, we observe that high-gating (leader) and low-gating (auxiliary) experts play distinct roles in MTL: leader experts dominate task-specific predictions, while auxiliary experts, despite their lower gating scores, often contain complementary knowledge that can enhance model performance. However, critical challenges persist: how to effectively identify and utilize the knowledge of the leader and auxiliary experts in a synergistic manner? To address this, we propose a novel Dynamic Experts Synergy (DES) mechanism that integrates Entropy-driven Experts Classification (EEC) and Multi-view Knowledge Recycle (MVKR). EEC dynamically partitions experts into leader and auxiliary groups by analyzing task-specific prediction and gating entropy, enabling adaptive allocation aligned with real-time task difficulty. MVKR effectively revisits knowledge from auxiliary experts through utility, diversity, and task-relatedness perspectives, ensuring comprehensive knowledge utilization. Extensive experiments on five datasets demonstrate the superiority of our DES against state-of-the-art methods.

CCS Concepts

• Information systems → Recommender systems; • Computing methodologies → Multi-task learning.

Keywords

Multi-task Learning; Recommender Systems; Entropy-driven Expert Classification; Multi-view Knowledge Recycle

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*Corresponding Authors



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1 Introduction

Recommender systems (RSs) are designed to filter, rank, and deliver personalized information aligned with user preferences reflected by user interactions [53]. In real-world applications, these interactions encompass various behavioral signals, like clicks, purchases, and shares, etc. Efficiently modeling and predicting these behaviors is essential for understanding user preferences and engagement. However, predicting each behavior in isolation can be inefficient. Thus, multi-task learning (MTL) has emerged as a promising approach to model these behaviors simultaneously [37, 38, 45, 50]. However, these signals often reflect different stages of the user engagement funnel and vary significantly in data sparsity and business importance, raising obstacles for MTL implementation. Specifically, dense tasks such as click-through rate (CTR) provide abundant feedback and drive core revenue, whereas sparse tasks such as predicting shares and post-view click-through & conversion rate (CTCVR), though less frequent, are critical to user engagement and long-term retention [38, 51]. This heterogeneity leads to varying optimization dynamics and even conflicts across tasks, introducing the well-known phenomenon of negative transfer [21, 32, 43, 63]. In this scenario, dominant and dense tasks like CTR overwhelm shared representations and degrade the learning performance of weaker and sparser tasks like sharing [40].

The expert-sharing pattern [37, 45, 49, 65], inspired by mixture-of-experts (MoE) [19], has emerged as an effective strategy to ease negative transfer by balancing inter-task correlations and differences, which typically appears in two broad forms: dense pattern and sparse pattern. Dense pattern [37] activates all experts and weights their outputs via input-generated gating. Although it may preserve more comprehensive knowledge, dense activation risks redundancy and overfitting [5]. Furthermore, as the distribution of gating scores tends to concentrate, experts with lower gating scores are often underutilized [37, 48, 64]. Sparse pattern engages only the most relevant experts via Top- k routing [16, 73]. Although it reduces redundancy by prioritizing high-gating experts, low-gating ones are directly excluded, potentially exacerbating the loss of valuable embedded information. Additionally, determining an appropriate k for different tasks poses a challenge due to variations

in their complexity. For verification, we conduct experiments on a naïve MME [37] with eight experts (dense pattern), combining it with different routing strategies (sparse pattern). Figure 1 presents results on AliExpress-FR and -US. Our observations include:

- (1) **The performance superiority between dense and sparse patterns depends on the selection of k , which exhibits dataset- and task-specific variability.** Sparse patterns can achieve comparable or even better performance in certain cases by focusing on a subset (k) of high-gating experts (leader experts). This underscores the importance of identifying leader experts to reduce redundancy. However, the optimal k varies across tasks and datasets.
- (2) **data-dense tasks often achieve better performance than data-sparse tasks when fewer experts are activated.** For example, when using the Top-1 sparse pattern, the performance decline relative to the dense pattern is substantially larger for the CTCVR task than for the CTR task. This suggests that data-sparse tasks such as CTCVR may require greater expert capacity to effectively leverage knowledge transferred from other tasks. The observation highlights a key limitation of fixed-capacity expert-sharing schemes in recommendation scenarios with pronounced disparities in task-specific data sparsity [33].
- (3) **Low-gating experts can act as performance enhancers for high-gating experts.** The performance gains from incorporating Bottom-1 into Top-2 are even better than those of Top-3 in some cases, indicating that low-gating experts (auxiliary experts) can also retain useful information, which may be overlooked by leader experts.

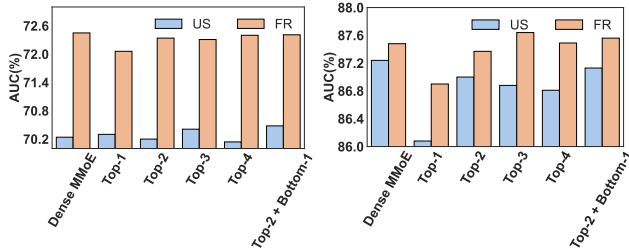


Figure 1: Comparison of dense and sparse patterns regarding CTR (left) and CTCVR (right).

Motivated by these insights, two key questions emerge: (1) *how to divide leader and auxiliary experts in a task-personalized manner*, and (2) *how to revisit the potential value embedded in auxiliary experts*. To this end, we propose Dynamic Experts Synergy (DES), which dynamically determines how many experts serve as leaders, making them contribute to specific tasks directly, and effectively revisit the knowledge of auxiliary experts via a multi-faceted approach to supplement the leader experts. DES consists of two steps: (1) **Entropy-driven Experts Classification (EEC)**: To dynamically determine the optimal number (k) of leader experts for each task, EEC operates on a core principle that is motivated by our empirical findings (as detailed in Section 3.1): aligning k co-directionally with real-time task difficulty. Specifically, a lower prediction entropy (indicating lower task uncertainty) or a lower gating entropy (reflecting more concentrated expert reliance) signifies a lower task difficulty, which in turn requires a smaller k .

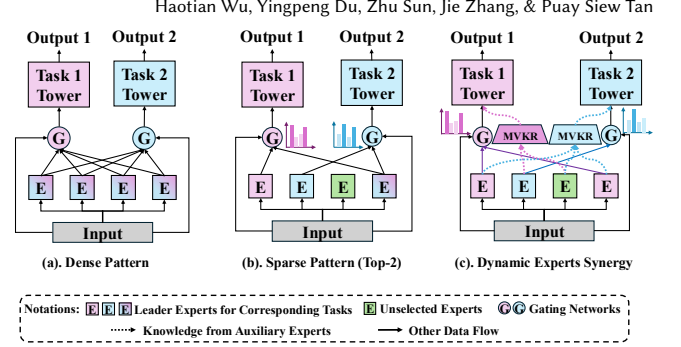


Figure 2: Comparison with dense and sparse expert sharing patterns. “E” represents expert.

To robustly implement this co-directional adjustment, EEC further employs a smooth dual-guard scheduler strategy, ensuring that adjustments to k occur only in response to statistically significant changes rather than instantaneously. This design not only aligns k with real-time task difficulty but also accounts for the timing of adjustments, preventing premature or excessive expert reallocation. (2) **Multi-view Knowledge Recycle (MVKR)**: To promote the use of auxiliary experts that can provide richer information, contribute to greater diversity, and better support specific tasks, we propose a multi-faceted approach to assess and mine the potential value of auxiliary experts. Specifically, MVKR revisits these auxiliary experts from three perspectives: (1) *Utility*: quantifying information abundance within each of them (intra-expert-level), (2) *Diversity*: assessing divergence of them from the aggregated leader experts (inter-expert-level), and (3) *Task-Relatedness*: measuring their alignment with a learnable task indicator (task-level). By fusing these views through a lightweight compressor module, MVKR augments leader experts with knowledge from auxiliary experts (Section 3.2).

Figure 2 contrasts DES with purely dense and sparse approaches. DES *dynamically selects* the appropriate sets of leader and auxiliary experts for each task according to its real-time task difficulty, then allows leader experts to contribute to tasks directly and reuses knowledge from auxiliary experts in a multi-faceted manner rather than eliminating them directly. This synergy strikes a balance: highlighting the dominant roles of leader experts and mitigating information loss of auxiliary experts. Our main contributions lie in three-fold: (1) We identify two critical limitations in existing expert-sharing MTL for RSs: the absence of a dynamic mechanism to partition leader and auxiliary experts and the underutilization of auxiliary experts. (2) We propose a Dynamic Expert Synergy (DES) mechanism, which adaptively partitions leader and auxiliary experts aligned with entropy-driven real-time task difficulty and effectively assesses and utilizes the potential value of auxiliary experts. (3) We conduct extensive experiments on five real-world datasets, showcasing that DES achieves state-of-the-art performance while offering enhanced flexibility and knowledge utilization in MTL.

2 Related Work

Multi-task Learning for RSs. MTL has been extensively used to predict diverse user feedback in RSs [58] through efficient data utilization and knowledge transfer, mainly classified into four groups: hard sharing, soft sharing, sparse sharing, and expert sharing [49]. Hard sharing employs shared bottom layers and task-specific top layers [3, 38, 55, 71], benefiting highly correlated tasks but often

causing negative transfer due to rigid parameter coupling. In contrast, soft sharing treats tasks separately, sharing information via correlation weights [39, 41] or attention mechanisms [7]. Though flexible, it struggles with efficiency and modeling inter-task correlations. Sparse sharing uses parameter masks to learn task-specific subnetworks within an over-parameterized base model [2, 10, 56], effectively reducing computational costs but often failing to capture fine-grained task nuances. Expert sharing, inspired by MoE [19], balances task correlation and differences by using experts for diverse knowledge and gating networks for customized expert knowledge aggregation [18, 22, 28, 34, 36, 37, 45, 47, 48, 57, 62, 72].

Some recent works [48, 64] have attempted to mitigate the polarization issues, that is the phenomenon of some experts being barely utilized for any tasks. BEnet [64] mitigates gate polarization by using a bagging layer and intra-expert attention to diversify feature usage, and HoME [48] applies expert normalization with Swish and a hierarchical mask to balance experts. Yet both mainly reduce low-gating experts to avoid collapse for MoE systems rather than extracting and utilizing their potentially valuable knowledge. Despite its advantages, dense expert sharing, which activates all experts for different tasks, faces redundancy. Thus, sparse expert sharing has emerged to resolve this with sparse gating strategies, such as top- k expert selection [6, 59, 73] or differentiable binary-encoded sparse gates [16]. However, selecting an optimal k remains challenging due to varying task- and data-specific needs. Moreover, both dense and sparse expert sharing suffer from the underutilization of low-gating experts. By contrast, our approach explicitly and dynamically separates leader and auxiliary experts and recycles auxiliary knowledge, turning underutilized experts into a complementary source instead of simply reducing them.

Sparse Mixture-of-Experts (SMoE) and Routing. Traditional MoE architectures employ input-dependent conditional computation to densely combine experts, enabling flexible modeling [19]. SMoE [42], pivotal in large language models [9, 20, 35], RSs [16, 73], and other domains [6, 11, 31, 68], improves efficiency by activating only a subset of experts via routing strategies. Typically, two main routing strategies are widely utilized: expert-choice (selecting top- k experts per token) [69, 70] and token-choice (assigning top- k tokens per expert) [8, 12, 26, 27]. However, the fixed k used in most methods limits the adaptability across tasks. As such, some recent work has explored the adjustable expert capacity by either the heuristic incremental search strategy [5, 67] or probabilistic value accumulation and truncation [17, 30, 60]. However, the former tends to converge to suboptimal solutions and is overly dependent on the initial choice of expert, and the latter encounters a dilemma wherein different tasks necessitate task-specific truncation thresholds. Thus, HyperMoE [66] introduces hypernetworks to utilize unselected experts in generative language models, mitigating performance loss from fixed k -selection. Despite prior advancements, our work makes distinct contributions by (1) adapting to the dynamic number of low-gating experts and (2) effectively leveraging low-gating experts in the MTL setting for more effective RSs.

3 Methodology

We introduce the proposed Dynamic Experts Synergy (DES) method, composed of two core components: Entropy-driven Experts Classification (EEC) and Multi-view Knowledge Recycle (MVKR). EEC

dynamically partitions the experts into leader and auxiliary experts, adapting to task-specific difficulty. MVKR further recycles knowledge from auxiliary experts to complement leader experts from three key perspectives: utility, diversity, and task-relatedness, enabling a synergistic knowledge integration mechanism that minimizes the loss of valuable information from auxiliary experts.

Task Definition. Let $\mathcal{S} = \{x_a, y_a, z_a\}_{a=1}^N$ is the observed dataset with N samples. x_a denotes the a -th sample with multiple fields (e.g. user field, item field), y_a and z_a denote the binary labels of click and conversion to the a -th sample respectively. $y = 1$ or $z = 1$ means click or conversion occurs respectively. $pCTR$ (post-view click-through rate), $pCTCVR$ (post-view click-through & conversion rate) are defined as follows:

$$pCTR = p(y = 1|x), pCTCVR = p(y = 1, z = 1|x). \quad (1)$$

Moreover, in the TenRec (QB-video) dataset [61], the objectives include predicting user interactions, i.e., follow, like, and share, in addition to click. Each of these tasks involves binary classification labels, which are defined in a similar way to the form of $pCTR$.

3.1 Entropy-driven Experts Classification (EEC)

Motivation and Overview. EEC aims to dynamically partition a set of M experts into *leader experts* (Top- k_j) and *auxiliary experts* ($M - k_j$) for each task j , based on the ranking of gating scores. This dynamic partitioning is driven by task-specific prediction entropy and gating entropy, ensuring adaptive expert allocation aligned with task difficulty and knowledge requirements. Prediction Entropy (U_j^t) reflects uncertainty in task predictions at iteration t [13, 25], where lower U_j^t implies higher confidence in predictions and requires a small number of experts. Gating Entropy (H_j^t) captures the concentration of gating scores [54] at iteration t , highlighting the reliance on a subset of experts. A higher H_j^t indicates a broader reliance on multiple experts, while a lower one suggests allowing focusing utilization of a smaller set of experts [17, 52].

To begin, we verify the intuition that k_j should vary in the same direction as U_j and H_j , that is, k_j increases when U_j and H_j rise and *vice versa*. Specifically, we track and average U_j and H_j across iterations. When the mean entropy drops below predefined thresholds, the *Normal Group* (*Norm*) reduces k_j while the *Contradiction Group* (*Con*) increases it. Results in Figure 3(a-b) show that *Norm* consistently outperforms *Con*, reinforcing the necessity of co-directionally aligning k_j with the trends of U_j and H_j . Furthermore, by introducing an artificial delay in the *Norm* group, we examine the effect of adjustment timing. Results indicate that a moderate delay in updating k_j enhances performance, underscoring the value of timing control over immediate adjustment in response to entropy shifts. Meanwhile, both entropies exhibit pronounced and consistent trends throughout training in Figure 3(c-d), which makes them reliable indicators to dynamically adjust expert allocation to match real-time task difficulty, with U_j reflecting task-specific performance and H_j capturing the intrinsic property of the MoE system.

Algorithmic Details. The EEC algorithm operates as follows. First, for each task j , the prediction entropy (U_j^t) and gating entropy (H_j^t)

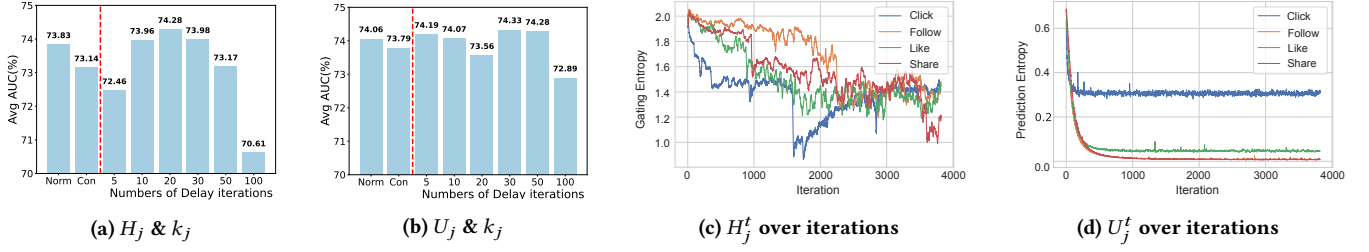


Figure 3: Results on TenRec, and similar trends can be noted on the other datasets.

at t -th iteration are computed as,

$$U_j^t = - \sum_c p_{j,c}^t \log(p_{j,c}^t + \epsilon), H_j^t = - \sum_{i=1}^M g_{j,i}^t \log(g_{j,i}^t + \epsilon), \quad (2)$$

where $p_{j,c}^t$ represents the predicted probability of class c for task j , $g_{j,i}^t$ is the gating probability for the i -th expert, and ϵ is a small constant for numerical stability.

Second, to avoid abrupt changes in k_j^t , we apply momentum mechanisms via exponential moving averages (EMAs). Specifically, we track $\Delta U_j^t = U_j^t - U_j^{t-1}$ and $\Delta H_j^t = H_j^t - H_j^{t-1}$, then update:

$$M_{U_j}^t = \alpha \Delta U_j^t + (1 - \alpha) M_{U_j}^{t-1}, M_{H_j}^t = \beta \Delta H_j^t + (1 - \beta) M_{H_j}^{t-1}, \quad (3)$$

where $\alpha, \beta \in [0, 1]$ are smoothing parameters. $M_{U_j}^t$ and $M_{H_j}^t$ thus represent smoothed estimates of recent changes in prediction entropy and gating entropy, respectively.

With the smoothed momentum signals ($M_{U_j}^t$ and $M_{H_j}^t$), a core challenge is to decide when to adjust the number of leader experts, k_j^t . Reacting to every minor fluctuation would introduce instability. To address this, we design a smooth dual-guard scheduler that triggers updates only in response to statistically significant and persistent changes. The *first-guard scheduler* decides whether the entropy momentum is significant enough to be transformed into a potential adjustment for k_j^t . We only proceed with an adjustment if the deviation in either entropy momentum is significant enough to exceed a standard deviation-scaled threshold (i.e., $|M_{U_j}^t - \mu_{M_{U_j}}^t| > \kappa \sigma_{M_{U_j}}^t$ or $|M_{H_j}^t - \mu_{M_{H_j}}^t| > \kappa \sigma_{M_{H_j}}^t$). Once this condition is met, we first synthesize the two momentum signals into a single, unified metric: the **difficulty score**, $\mathcal{D}_j^t$. This score functions as an intermediate signal that initiates a potential adjustment to k_j^t : its sign dictates the adjustment's direction (increase or decrease), while its magnitude influences the extent of the proposed potential adjustment. The difficulty score is computed as a weighted sum of the momentum of the changes in entropies:

$$\mathcal{D}_j^t = M_{U_j}^t \times w_{U_j}^t + M_{H_j}^t \times w_{H_j}^t. \quad (4)$$

To adaptively determine the dominant signal at iteration t , we assign higher weights to the momentum that exhibits greater deviation, ensuring the difficulty score precisely reflects real-time task difficulty. Specifically, the weights $w_{U_j}^t$ and $w_{H_j}^t$ are computed as:

$$Q_{U_j}^t = \frac{|M_{U_j}^t - \mu_{M_{U_j}}^t|}{\sigma_{M_{U_j}}^t}, Q_{H_j}^t = \frac{|M_{H_j}^t - \mu_{M_{H_j}}^t|}{\sigma_{M_{H_j}}^t}, \quad (5)$$

$$w_{U_j}^t, w_{H_j}^t = \text{Softmax}(Q_{U_j}^t, Q_{H_j}^t). \quad (6)$$

This mechanism ensures that the entropy with a significant deviation exerts greater influence on the difficulty score, aligning it with task dynamics.

To ensure k_j^t adapts responsively to task difficulty while avoiding unnecessary fluctuations, we design a scaling mechanism that amplifies significant entropy changes while suppressing minor variations. Specifically, the difficulty score $\mathcal{D}_j^t$ is mapped to the potential adjustment Δ_j^t in k_j^t via:

$$\Delta_j^t = \text{sign}(\mathcal{D}_j^t) \left[\exp(\gamma |\mathcal{D}_j^t|^\delta) - 1 \right], \quad (7)$$

where $\text{sign}(\mathcal{D}_j^t)$ determines the direction of adjustment, and $\gamma, \delta > 0$ control the sensitivity to entropy deviations. When $|\mathcal{D}_j^t| \rightarrow 0$, the adjustment remains minimal, preventing unnecessary updates. As it increases, the exponential term grows rapidly, enabling faster adaptation to substantial entropy shifts.

To prevent frequent oscillations in k_j^t , we accumulate the increment Δ_j^t at each iteration into an accumulator $\mathcal{A}_j^t$. This process ensures that small, transient potential adjustments do not immediately alter k_j^t . Once $|\mathcal{A}_j^t - \mu_{\mathcal{A}_j}^t|$ surpasses a standard deviation-scaled bound (*second-guard scheduler*, $|\mathcal{A}_j^t - \mu_{\mathcal{A}_j}^t| > \theta \sigma_{\mathcal{A}_j}^t$, which decides whether this accumulated potential adjustment is significant enough to be executed as an actual adjustment to k_j^t), indicating a persistent or significant deviation, k_j^t is updated accordingly:

$$k_j^t = k_j^{t-1} + \text{sign}(\mathcal{A}_j^t), \mathcal{A}_j^t = \mathcal{A}_j^{t-1} + \Delta_j^t. \quad (8)$$

Accordingly, a steady decline in entropy leads to a gradual reduction in leader experts, while rising entropy triggers the reintroduction of additional leaders to meet evolving task difficulties. This mechanism allows k_j^t to respond more robustly to genuine shifts in task confidence and gating distributions while filtering out minor fluctuations. Finally, the k_j^t experts with the highest gating scores are designated as leader experts, which directly contribute to specific tasks, while the remaining experts are categorized as auxiliary experts. Then the auxiliary experts can be revisited in subsequent stages. The pseudo-code of EEC is shown in Appendix D.

3.2 Multi-view Knowledge Recycle (MVKR)

Unlike conventional Top- k routing, which discards the knowledge of unselected experts, DES leverages auxiliary experts to enhance task-specific modeling through MVKR. To effectively integrate their knowledge, we assess auxiliary experts from three levels: intra-expert (utility), inter-expert (diversity), and task-level (task-relatedness). These perspectives are fused into a multi-view representation and incorporated into leader experts via a lightweight

compression network. As shown in Figure 4, the outputs of auxiliary experts are fused in a multifaceted manner into three fixed-dimensional representations. This setup ensures that the potentially valuable knowledge embedded in them can still be smoothly utilized despite the dynamic number of auxiliary experts.

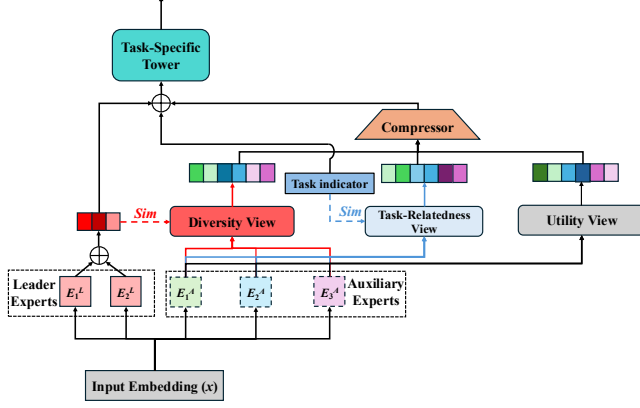


Figure 4: The illustration of MVKR.

Utility View. At the intra-expert level, we quantify the utility of each auxiliary expert by introducing *information abundance* [14], which measures the uniformity of singular value distribution [23]. A higher value indicates more evenly distributed singular values and reduced dimensional collapse. Given a matrix E , its information abundance is defined as [14]:

$$\phi(E) = \frac{\|\sigma\|_1}{\|\sigma\|_\infty}, \quad (9)$$

where $\|\sigma\|_1$ and $\|\sigma\|_\infty$ are the L_1 and L_∞ norm of singular value vector, respectively. For each auxiliary expert, its knowledge can be fused by:

$$Q_j^{uti} = \sum_{i=1}^{M-k_j} \alpha_{i,j}^{uti} E_i^A(x), \quad \alpha_{i,j}^{uti} = \frac{\exp(\phi(E_i^A(x)))}{\sum_{i=1}^{M-k_j} \exp(\phi(E_i^A(x)))}, \quad (10)$$

where $E_i^A(x)$ is the output of i -th auxiliary expert and x is input embedding for current batch. In this way, auxiliary experts with higher information abundance will be promoted.

Diversity View. At the inter-expert level, we encourage diverse knowledge contribution by assigning greater weights to auxiliary experts whose outputs exhibit higher deviation from those of the leader experts. Specifically, we first obtain the aggregated representation of the leader experts g_j^L by,

$$g_j^L = \sum_{i=1}^M G(x)_j E_i(x), \quad G(x)_j = \text{Topk}(\text{Softmax}(xW_j^g)), \quad (11)$$

where $G(\cdot)_j$ is the routing network for task j , Topk selects the $\text{Top-}k_j$ (renormalization) leader experts with highest gating score, k_j is dynamically determined by Equation 8 in EEC, and W_j^g is a trainable parameter matrix. Then, the diversity score (ρ_i^j) between outputs of each auxiliary expert i and g_j^L is computed, and auxiliary experts' knowledge is fused as below,

$$\rho_i^j = 1 - \frac{E_i^A(x) \cdot g_j^L}{\|E_i^A(x)\| \|g_j^L\| + \epsilon}, \quad Q_j^{div} = \sum_{i=1}^{M-k_j} \alpha_{i,j}^{div} E_i^A(x), \quad (12)$$

$$\alpha_{i,j}^{div} = \frac{\exp \rho_i^j}{\sum_{i=1}^{M-k_j} \exp \rho_i^j}. \quad (13)$$

Task-Relatedness View. At the task level, we aim to ensure that auxiliary expert knowledge is effectively aligned with the specific task requirements. While informativeness and diversity help retain valuable and complementary knowledge, they do not guarantee that the retained knowledge is relevant to the target task. To address this, we measure the alignment of auxiliary expert knowledge with the task by evaluating its similarity to a task indicator ζ , which represents task-specific characteristics [46]. Specifically, the task indicator is a randomly initialized learnable vector maintained for each task and is fed into task-specific towers to capture task-specific information. The task-relatedness score $\alpha_{i,j}^{rel}$ is computed, and auxiliary experts' knowledge is fused as below,

$$Q_j^{rel} = \sum_{i=1}^{M-k_j} \alpha_{i,j}^{rel} E_i^A(x), \quad \alpha_{i,j}^{rel} = \frac{\exp(E_i^A(x) \cdot \zeta_j)}{\sum_{i=1}^{M-k_j} \exp(E_i^A(x) \cdot \zeta_j)}. \quad (14)$$

This encourages auxiliary experts with higher task-relatedness to contribute more significantly.

Finally, the knowledge from the three views is concatenated and fed to a compression network (MLP) $h_j(\cdot)$, yielding auxiliary experts' knowledge g_j^A , with the same dimensionality as g_j^L ,

$$g_j^A = h_j(Q_j), \quad Q_j = \text{Concatenate}(Q_j^{uti}, Q_j^{div}, Q_j^{rel}). \quad (15)$$

Accordingly, g_j^A and g_j^L are aggregated and fed into task-specific towers to complete the prediction.

4 Experiments

4.1 Experimental Setup

Five datasets are adopted in the experiments, whose statistics are shown in Appendix Table 6. Amongst them, AliExpress-NL, AliExpress-ES, AliExpress-US, and AliExpress-FR are collected from real-world traffic logs of the AliExpress search system from the Netherlands, Spain, the US, and France, respectively [29]. The TenRec (QB-Video) dataset is collected from the QQ BOW platform, which contains diverse user interactions, i.e., clicks, likes, shares, and follows [61]. Following the mainstream MTL-based RS research [44, 65], we randomly split TenRec (QB-Video) datasets into training, validation, and test sets with a ratio of 8:1:1. We randomly split all the AliExpress-series datasets into training, validation, and test sets with a ratio of 4:1:1. We adopt AUC as the evaluation metric. To verify the effectiveness of our proposed DES, we compare with 13 competitive baselines: (1) **Hard sharing pattern:** *Shared-Bottom* [3], *AITM* [55]; (2) **Soft sharing pattern:** *Cross-Stitch* [39]; (3) **Dense expert sharing pattern:** *OMoE* [19], *MMoE* [37], *PLE* [45], *MetaHeac* [72], *AdaTT* [28], *STEM* [44], *HoME* [48]; and (4) **Sparse experts sharing pattern:** *Naïve Top-k* [42], *Dselect-k* [16], *Dynamic-MoE* [17]. More details for baselines are shown in Appendix A. Moreover, detailed hyperparameter settings are in Appendix B.

4.2 Experimental Results and Analysis

Overall Comparison. The performance of all methods on the five datasets is presented in Tables 1-2, where the best results are in bold, and the runner-up is underlined. Several observations are noted: **First**, our DES achieves the best average AUC (Avg AUC) across five datasets. Specifically, taking the TenRec (QB-Video) dataset as an example, HoME, a recent state-of-the-art expert-sharing method, achieves about a 0.17% relative improvement in Avg AUC over STEM, while our DES yields about a 0.59% relative improvement over HoME. Compared with PLE, a classical competitive expert-sharing baseline, DES attains about a 1.54% relative improvement in Avg AUC. Furthermore, the performance superiority of DES on the CTR task is more significant on the AliExpress-NL, US, and FR datasets, with larger scales and more features. **Second**, on the TenRec (QB-Video) dataset, most of the hard-sharing, soft-sharing, and expert-sharing patterns suffer from different degrees of negative transfer phenomena for the task ‘Like’, which has fewer positive feedback signals. DES effectively improves such a situation, which demonstrates that DES enhances the effectiveness of tasks in utilizing each other’s knowledge by facilitating collaboration between leader experts and auxiliary experts, thereby ensuring that data-sparse tasks can still benefit from other tasks. In addition, TenRec (QB-Video) has less user- and item-side information and thus relies more on task-to-task knowledge transfer. The advanced Avg AUC of DES — showing around a 2.05% relative gain over single-task learning — further validates that our DES effectively facilitates mutual benefits among tasks. **Third**, DES’s consistently superior performance over HoME demonstrates that reasonably recycling the knowledge embedded in low-gating experts outperforms simply reducing or suppressing them.

Table 1: AUC (%) comparison on TenRec (QB-Video).

Model	Click	Follow	Like	Share	Avg
Single Task	91.46	65.32	70.07	67.89	73.69
Shared Bottom	89.32	66.82	68.33	67.25	72.93
AITM	89.48	66.96	69.45	68.73	73.66
Cross-Stitch	90.12	67.58	69.76	67.69	73.79
OMoE	90.35	66.48	69.54	68.68	73.76
MMoE	90.08	67.65	69.72	69.03	74.12
PLE	89.98	<u>68.74</u>	69.83	67.70	74.06
MetaHeac	90.26	66.89	70.11	66.84	73.53
AdaTT	91.84	67.82	69.73	67.99	74.35
STEM	91.39	68.32	<u>70.17</u>	68.64	74.63
HoME	<u>91.91</u>	68.56	70.05	68.52	<u>74.76</u>
Naïve Top- k	90.38	67.88	69.62	68.09	73.99
DSelect- k	91.63	67.61	70.01	68.41	74.42
Dynamic-MoE	90.87	68.26	69.86	68.04	74.26
DES	91.96	69.46	70.65	<u>68.74</u>	75.20

In terms of routing strategies, Sparse MoE (Naïve Top- k , Dselect- k , Dynamic-MoE) exhibits comparable performance to Dense MoE (MMoE), where underutilization [48] (i.e., polarization [37, 64]) and overfitting [67] may be the main reasons for such a phenomenon. Such a finding aligns with prior studies [6, 15]. DES illustrates that

dynamically adjusting leader experts based on real-time task difficulty (via EEC) is more effective than either fixing k globally or adjusting expert capacity based on a predefined cumulative gating threshold (Dynamic-MoE) [17]. Meanwhile, auxiliary-expert reuse (through MVKR) provides a crucial advantage in harnessing auxiliary experts’ knowledge, an aspect that conventional sparse or dense solutions often overlook.

Ablation Study. We now analyze the effect of essential components of DES, including prediction entropy (PE), gating entropy (GE), and the Accumulator in EEC, as well as MVKR. Specifically, we individually remove each component (i.e., w/o GE, w/o PE, w/o Accumulator and w/o MVKR) and measure how the model’s performance changes across the four tasks.

Table 3: AUC (%) comparison on components of DES.

	Click	Follow	Like	Share	Avg
DES w/o MVKR	91.46	68.02	69.84	67.98	74.33
EEC w/o GE	90.87	69.04	69.91	67.89	74.43
EEC w/o PE	91.97	67.85	70.19	68.52	74.63
EEC w/o Accumulator	91.31	68.38	69.72	68.47	74.47
DES	91.96	69.46	70.65	68.74	75.20

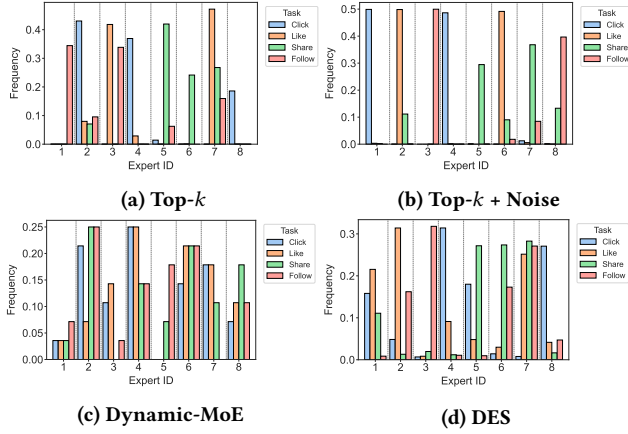
Table 3 shows the performance (AUC) where three major insights are gained. **First**, removing MVKR results in the most significant performance drop, highlighting (a) the effectiveness of MVKR in enhancing the knowledge transfer for task-specific modelling and (b) the importance of recycling knowledge from auxiliary experts. **Second**, excluding gating entropy (GE) or prediction entropy (PE), both lead to performance degradation, indicating that both entropy signals are crucial for dynamic expert division. Notably, removing GE has a greater impact, suggesting that the gating entropy better reflects real-time task difficulty in the TenRec (QB-Video) dataset. **Third**, eliminating the accumulator reduces performance, underscoring the necessity of long-range momentum accumulation for stabilizing k -adjustment. Overall, these findings confirm that *each component (i.e., MVKR, GE, PE, and the accumulator) plays a critical role in DES’s effectiveness, reinforcing its design for dynamic expert division and auxiliary knowledge reuse.*

Expert Utilization Analysis. To understand how various expert-sharing patterns (MMoE with Top- k routing, Top- k with Gaussian Noise [6, 42], Dynamic-MoE, and our DES) distribute “expert usage” across different tasks, we track and record the frequency with which different tasks activate each expert as a “leader” during training. The results are shown in Figure 5.

In Top- k , certain experts are heavily activated by a single task while remaining underutilized elsewhere, suggesting that a fixed k can reinforce high-gating experts at the expense of more balanced utilization. For example, Experts 1, 6, 8 are only activated by “Follow”, “Share”, “Click” tasks, respectively. In addition, the “Click” task only activated Expert 8 less than 20% of the time, indicating the overall low utilization of Expert 8, namely load imbalance issue [42]. Although the introduction of learnable Gaussian noise mitigates the load imbalance problem to some extent [6], it creates an over-specialization problem that may weaken the modeling of task correlation [6]. For example, the “Click” task, almost exclusively activates Expert 1 or Expert 4, while the “Like” task, almost

Table 2: AUC (%) comparison on AliExpress-US, AliExpress-NL, AliExpress-FR, AliExpress-ES.

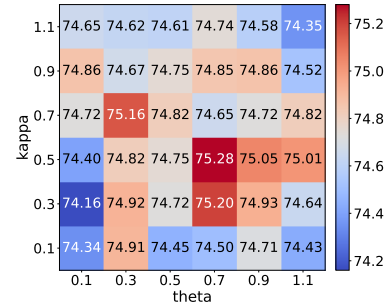
Methods	AliExpress-US		AliExpress-NL		AliExpress-FR		AliExpress-ES	
	CTR AUC	CTCVR AUC	CTR AUC	CTCVR AUC	CTR AUC	CTCVR AUC	CTR AUC	CTCVR AUC
Single-Task	70.52	86.44	72.58	85.17	72.28	87.36	72.41	87.67
Shared-Bottom	70.59	86.90	72.32	85.65	72.44	87.09	72.50	87.24
AITM	70.77	87.36	72.11	86.21	72.30	87.62	72.47	88.02
Cross-Stitch	70.71	86.82	72.14	85.37	72.43	87.23	72.33	87.37
OMoE	70.52	86.60	72.31	85.71	72.51	87.33	72.58	87.81
MMoE	70.24	87.24	72.61	86.07	72.45	87.48	72.76	88.06
PLE	70.65	87.22	72.42	85.93	72.26	87.65	72.60	88.37
MetaHeac	70.89	86.67	72.41	85.73	72.63	86.95	72.23	87.58
AdaTT	70.48	87.41	72.57	86.01	72.48	<u>87.73</u>	72.64	87.75
STEM	<u>71.01</u>	<u>87.54</u>	72.50	85.86	72.46	87.32	72.68	88.13
HoME	70.42	87.46	<u>72.75</u>	86.14	72.61	87.48	72.74	<u>88.48</u>
Naïve Top- k	70.20	87.00	72.26	86.10	72.34	87.37	72.59	88.23
DSelect- k	70.70	87.13	72.56	<u>86.23</u>	72.45	87.07	<u>72.78</u>	88.25
Dynamic-MoE	70.57	87.39	72.35	85.54	<u>72.68</u>	87.39	72.45	88.31
DES	72.10	87.55	73.47	86.47	73.47	87.77	73.11	88.91

**Figure 5: Illustration of the frequency of experts who are selected as leaders for different tasks.**

exclusively activates Expert 2 and Expert 6, which suggests that the correlation between the “Click” and “Like” task is difficult to model adequately.

Conversely, Dynamic-MoE yields a more uniform but less specialized distribution—most experts see moderate but shallow engagement across tasks, potentially inhibiting the deep specialization desired in the expert-sharing pattern [5, 28]. Compared to Top- k and Dynamic-MoE, DES achieves a more balanced (no experts are selected as leaders by all tasks with very low frequency) yet selectively focal distribution of expert usage. More importantly, DES largely insulates the model from load imbalance. This is because even if some of the experts are assigned lower gating scores, they are still able to contribute incremental information to a specific task through MVKR and stimulate the full utilization of different experts to a greater extent.

Effect of Thresholds κ and θ . To investigate how thresholds κ and θ , two core hyperparameters in the smooth dual-guard scheduler, impact the performance of DES, we apply a grid search in the range of $[0.1, 1.1]$ stepped by 0.2 for both thresholds. Figure 6 shows the average AUC across the four tasks on TenRec (QB-Video) regarding different settings of κ and θ .

**Figure 6: Effect of κ and θ on Avg AUC (%) achieved by DES on TenRec (QB-Video).**

More figures showing AUC on each individual task are shown in Appendix E. Several observations are noted. **First**, a clear “high-center, low-edge” structure emerges. Specifically, the region around $\kappa \in [0.3, 0.9]$ and $\theta \in [0.3, 0.9]$ not only achieves high accuracy (at or above 75.0%) but also exhibits comparatively stable values across neighboring cells, suggesting that moderate thresholds in both momentum-based release and accumulator-based adjustments yield a robust “sweet spot”. In comparison, in most cases, notable performance dropping toward 74.2–74.4% appears at extreme value settings for κ and θ , suggesting that overly strict or lenient release conditions undermine the model’s ability to adaptively select experts. This broad optimal zone makes parameter selection more flexible than precisely tuning a discrete k in Naïve Top- k routing.

We also notice that most of the (κ, θ) combinations tested outperform the Top- k baseline in our paper. **Second**, the hyperparameter κ exhibits overall greater sensitivity compared to θ . This is evident from the larger fluctuations in performance values along the vertical axis (i.e., across different κ values) for fixed θ .

Effect of Different Views of Knowledge. To examine how each view of auxiliary knowledge, i.e., Utility, Diversity, Task-Relatedness (abbreviated as U, D, R), contributes to the overall performance, we test different combinations of views in MVKR across the four tasks on TenRec (QB-Video), shown in Table 4.

Table 4: Effect of different views of knowledge.

Knowledge View			Click	Follow	Like	Share	Avg
U	R	D					
✓			91.76	68.33	70.19	67.79	74.52
	✓		90.92	66.85	69.68	68.21	73.92
		✓	91.69	68.17	69.66	68.02	74.39
✓	✓		92.01	69.09	69.74	68.65	74.87
✓		✓	91.86	68.02	69.83	67.94	74.41
	✓	✓	91.92	68.70	70.01	67.78	74.60
✓	✓	✓	91.96	69.46	70.65	68.74	75.20

First, the "U" variant outperforms other single-view variants ("R" and "D"), suggesting that it contains the most individually valuable information among the three. **Second**, "U+D" combination yields the worst performance among all dual-view variants, aligning with the persistently higher similarity among auxiliary experts compared to "U+R" (the best dual-view variant) during training as shown in Figure 7(a). This suggests that auxiliary experts exhibit substantial overlap under the "U+D" view, thereby limiting the completeness of the knowledge they contribute. **Third**, "U+D+R" combination outperforms all other variants, demonstrating its ability to effectively leverage complementary knowledge from auxiliary experts to enhance the leader experts. This is further supported by the increased similarity gap between leader and auxiliary experts under the "U+D+R" views compared to the variant without any view integration throughout training, as shown in Figure 7(b).

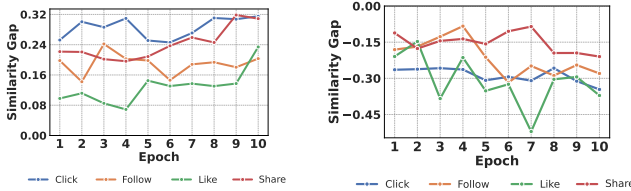


Figure 7: (a) The similarity gaps among auxiliary experts under "U+D" (the worst variant with dual-view) compared to "U+R" (the best variant with dual-view); (b) The similarity gap between leader and auxiliary experts under "U+D+R" compared to the variant without integration of any view.

Generalizability of DES. To further assess the generalizability of DES beyond RSs, we evaluate it on MultiMNIST [4], a widely-used MTL dataset in the computer vision (CV) community. MultiMNIST consists of 32×32 pixel images containing two handwritten digits positioned at the top-left and bottom-right corners, requiring simultaneous classification of both digits (Tasks 1 and 2). Across this

dataset, DES consistently surpasses all selected baselines, achieving the highest average accuracy while maintaining superior task balance. This result is particularly compelling because CV tasks emphasize spatially hierarchical features and local pattern recognition, which differ fundamentally from the heterogeneous user-item interactions and sparse categorical signals in RSs. The ability of DES's dynamic expert selection mechanism to adapt effectively to both dense visual representations and sparse behavioral data demonstrates its versatility across diverse MTL settings.

Table 5: Accuracy (%) comparison on MultiMNIST dataset.

Method	Task 1 Acc.	Task 2 Acc.	Avg. Acc.
Single-Task	93.49	91.98	92.74
MMoE	93.53	92.25	92.89
PLE	93.39	91.96	92.68
HoME	93.46	92.16	92.81
DES	94.33	92.94	93.64

5 Conclusions

We propose Dynamic Experts Synergy (DES), a novel framework that fundamentally addresses the rigid expert allocation and under-utilization of auxiliary experts in the MTL setting for RSs. Specifically, DES is equipped with two core components, namely, Entropy-driven Experts Classification (EEC) and Multi-view Knowledge Recycle (MVKR). EEC dynamically adapts leader and auxiliary experts allocation by jointly modeling task-specific prediction entropy and gating entropy, enabling real-time alignment with task complexity; while MVKR pioneers a multi-faceted approach to harness auxiliary experts through utility, diversity, and relatedness views, mitigating knowledge loss in auxiliary experts. Experiments across five real-world datasets validate that DES achieves state-of-the-art performance, particularly enhancing data-sparse tasks by synergizing the knowledge of the leader and auxiliary experts. In addition, DES minimizes the impact of load imbalances as no experts are directly eliminated, and balances comprehensiveness and specialization.

Limitations and Future Directions. While DES introduces computational overhead, with a runtime of 503 s/epoch (compared to AdaTT's 378 s/epoch and STEM's 412 s/epoch) on AliExpress-ES (the largest dataset among the five), its memory usage is lower at 36 MB/batch, compared to 45 MB/batch for both AdaTT and STEM. Such insights yield promising directions for future improvements: more lightweight expert allocation and knowledge recycling strategies to reduce the computational overhead and better adapt to large-scale application scenarios. Meanwhile, when the hidden dimension of the experts is large, approximate SVD can be employed to obtain singular values for information abundance, thereby further reducing computational complexity.

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A More Details of Adopted Baselines.

Here is a more detailed description of the baseline used in this paper.

- (1) **Single Task**: Trained MLP model for each task individually. In implementation, it is essentially equivalent to a shared bottom model with only one tower.
- (2) **Shared Bottom** [3]: The classic example of hard sharing pattern. All tasks share a feature extractor at the bottom to capture shared knowledge. A task-specific tower is set at the top of each task to extract task-specific knowledge.
- (3) **AITM** [55]: Improved shared bottom model by introducing adaptive information transfer modules between the outputs of different tasks.
- (4) **Cross-Stitch** [39]: The classic example of soft sharing, which uses learnable cross-stitch units to share knowledge between two tasks.
- (5) **OMoE** [19]: dense expert-sharing method, which shares one gate among tasks.
- (6) **MMoE** [37]: The classic example of the dense expert-sharing method, which sets an individual gate for each task to capture differences among tasks.
- (7) **PLE** [45]: The classic example of the dense expert-sharing method, which explicitly divides shared and task-specific experts based on MMoE.
- (8) **MetaHeac** [72]: Dense expert-sharing method, which employs meta-learning to capture the relationship among tasks.
- (9) **AdaTT** [28]: Dense expert-sharing method, which consists of multi-level task-specific and optional shared fusion units.
- (10) **STEM** [44]: MoE-based MTL method, which introduces task-specific and shared input embeddings based on PLE.
- (11) **HoME** [48]: MoE-based MTL method, which employs a hierarchical mask with expert normalization and Swish activation to balance expert outputs and mitigate collapse.
- (12) **Naïve Top- k** [42]: The classic example of the sparse expert-sharing methods, which selects experts with the largest k gate values. In this paper, to balance the performance and sparsity, k is set to 2.
- (13) **DSelect-K** [16]: A continuously differentiable sparse gating method for the MoE architecture.
- (14) **Dynamic-MoE** [17]: Set a threshold P to truncate when the gating value accumulates up to P and select the number of corresponding experts. (The P is set to 0.5.)

B Hyper-parameters Setting

To ensure fair comparison, we adopt Adam [24] with the initial learning rate $1e^{-3}$ to optimize all the models. The embedding size of the input is 128 for AliExpress-ES, US, NL, FR, and 16 for TenRec(QB-Video). We adopt two-layer MLP networks with ReLU activation in the shared bottom, task towers, and experts. The hidden layer sizes are [512,256], [128,64], and [64,32], respectively. The total number of experts is set to 8 for all the expert-sharing models. “ k_{max} ” is also set to 8 for DES. We tune other hyperparameters by Optuna [1], with the hyper-parameters search space specified as follows: $\alpha, \beta \in \{0.1, 0.2, \dots, 0.9\}$, $\kappa \in \{0.1, 0.3, \dots, 1.1\}$, $\theta \in \{0.1, 0.3, \dots, 1.3\}$, $\gamma \in \{1e^{-3}, 3e^{-3}, 5e^{-3}, 7e^{-3}, 9e^{-3}, 1e^{-2}\}$, $\delta \in \{0.1, 0.2, \dots, 1.0\}$. For

TenRec (QB-Video) dataset, α and β are set to 0.7, 0.3, respectively, γ and δ are set to 0.001 and 0.8, respectively, κ and θ are set to 0.3, 0.7, respectively. In terms of the four AliExpress datasets, the $\alpha, \beta, \gamma, \delta$ are set to 0.3, 0.6, 0.007, and 0.5, respectively. The setting of κ and θ , considering their importance in controlling the smooth dual-guard scheduler, varies across four datasets: AliExpress-NL ($\kappa = 0.9, \theta = 0.9$), AliExpress-US ($\kappa = 1.1, \theta = 0.5$), AliExpress-FR ($\kappa = 0.9, \theta = 0.9$), AliExpress-ES ($\kappa = 0.3, \theta = 0.3$). We adopt cross-entropy as the loss function for each task, following prior studies [37, 55], the weights for different tasks’ losses are all set to 1, indicating that tasks are treated equally. In addition, to ensure training stability, we add a gating entropy regularization constraint to DES’s loss with a weight of 0.001 on the TenRec (QB-Video). All the experiments are run on RTX 4090.

C Statistics of experimental datasets.

Table 6: Statistics of experimental datasets.

Datasets	#item	#impression	#click	#conversion	#like	#share	#follow
AliExpress-US	8M	27.4M	450K	10.9K	-	-	-
AliExpress-ES	8.7M	31.6M	841K	19.1K	-	-	-
AliExpress-NL	6M	17.7M	382K	13.8K	-	-	-
AliExpress-FR	7.4M	27.4M	535K	14.4K	-	-	-
TenRec(QB-Video)	130K	2.0M	1.7M	-	20.7K	2.5K	2.5K

D Pseudo-Code of EEC

Algorithm 1 Entropy-driven Experts Classification (EEC)

Require: M experts $\{E_i\}_{i=1}^M$, routing networks G_j for tasks $j = 1, \dots, T$
Ensure: For each j , a dynamic k_j partitioning experts into Leader (top- k_j) and Auxiliary ($M - k_j$)

```

1: for  $j \leftarrow 1$  to  $T$  do
2:   Init:  $k_j^t \leftarrow k_{max}$ ;  $U_j^t, H_j^t \leftarrow 0$ ;  $\mu_{M_{U_j}^t}, \mu_{M_{H_j}^t}, \mu_{\mathcal{A}_j^t} \leftarrow 0$ ;
    $\sigma_{M_{U_j}^t}, \sigma_{M_{H_j}^t}, \sigma_{\mathcal{A}_j^t} \leftarrow 1$ 
3:   for  $t \leftarrow 1$  to  $N$  do
4:     Compute  $U_j^t, H_j^t$  (Eq. 2); update  $M_{U_j}^t, M_{H_j}^t$  (Eq. 3)
5:     Update  $\mu_{M_{U_j}^t}, \mu_{M_{H_j}^t}, \sigma_{M_{U_j}^t}, \sigma_{M_{H_j}^t}$  via EMA
6:     if  $|M_{U_j}^t - \mu_{M_{U_j}^t}| > \kappa \sigma_{M_{U_j}^t}$  or  $|M_{H_j}^t - \mu_{M_{H_j}^t}| > \kappa \sigma_{M_{H_j}^t}$  then
7:       Compute difficulty  $\mathcal{D}_j^t$  (Eq. 4); map to  $\Delta_j^t$  (Eq. 7)
8:       Update accumulator  $\mathcal{A}_j^t$ , then update  $\mu_{\mathcal{A}_j^t}, \sigma_{\mathcal{A}_j^t}$  via EMA
9:       if  $|\mathcal{A}_j^t - \mu_{\mathcal{A}_j^t}| > \theta \sigma_{\mathcal{A}_j^t}$  then
10:         $k_j^t \leftarrow$  update rule (Eq. 8)
11:       else
12:         $k_j^t \leftarrow k_j^{t-1}$ 
13:       end if
14:     end if
15:     Assign Top- $k_j^t$  as Leader Experts by routing networks; rest as
       Auxiliary Experts, then processing by MVKR
16:   end for
17: end for

```

(Note: We set upper and lower bounds for k_j^t (8 and 2, respectively) to prevent over-adjustment).

E Additional details of ablation study on the thresholds

Figure 8 presents three heatmaps for the tasks Click, Like, and Share. Click (with much higher positive feedback quantity than Like and Share tasks) achieves over 91% AUC across a relatively broad (κ , θ) range, implying that a larger volume of positive feedback allows the model to remain robust under various threshold settings. In contrast, Like and Share show more pronounced changes when κ or θ becomes too large or too small. This pattern suggests that sparse tasks then rely heavily on multi-task synergy: an appropriately tuned dynamic threshold (particularly near middle values of κ and θ) helps compensate for limited task-specific signals, allowing information to be shared effectively from tasks with richer data. Furthermore, TenRec’s smaller feature numbers likely amplify the importance of cross-task knowledge, since each individual task may not have enough features or volume on its own to train a stable model.

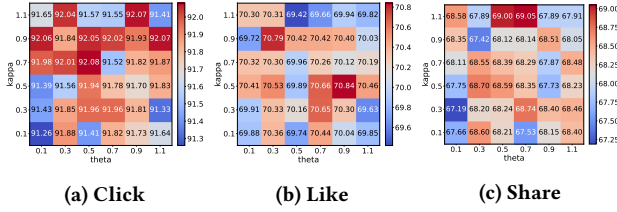


Figure 8: Effect of κ and θ on DES’s performance on TenRec (QB-Video) dataset

Figures 9– 12 illustrate two tasks (CTR AUC, and CTCVR AUC, Avg AUC denotes the Average AUC of CTR and CTCVR tasks) evaluated on significantly larger AliExpress datasets. Compared to the smaller-scale TenRec (QB-Video) results, these heatmaps show broader high-performance regions for moderate values of κ and θ . This observation suggests that with richer data and more features, the model can better benefit from multi-task synergy and features themselves without requiring extremely narrow threshold settings. However, extreme threshold settings (e.g., very low or very high κ or θ) may still cause notable drops in AUC, indicating that overly frequent or excessively delayed adjustments of k can degrade the shared representation learned across tasks. Overall, these findings demonstrate that appropriately balancing κ and θ remains crucial in large-scale environments, ensuring that the proposed DES can effectively align to both abundant data and cross-task complementarities.

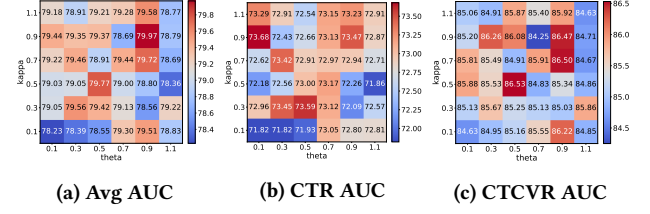


Figure 9: Effect of two thresholds on DES’s performance on AliExpress-NL dataset

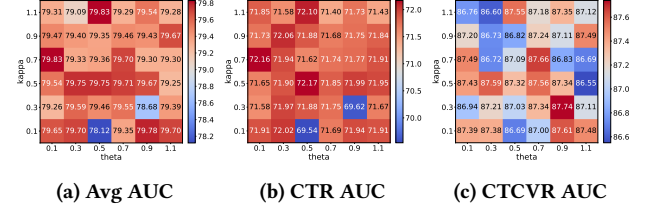


Figure 10: Effect of two thresholds on DES’s performance on AliExpress-US dataset

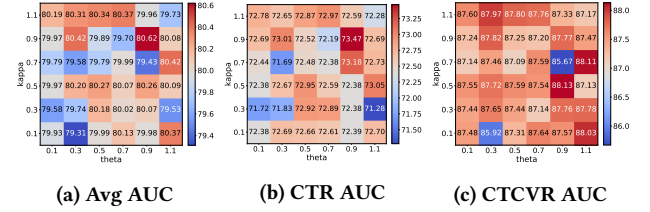


Figure 11: Effect of two thresholds on DES’s performance on AliExpress-FR dataset

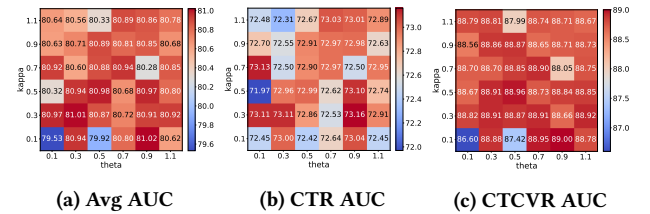


Figure 12: Effect of two thresholds on DES’s performance on AliExpress-ES dataset